

Improving User-Centric Clustering for Near-Field Cell-Free XL-MIMO

Why current rules look the same, which metrics prove a clustering is better, and four proposed strategies

(Version 2: claims checked against the full texts of the cited papers)

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Summary in five lines

1. In your code, the channel-norm (CN) and information-rate (IR) clusters are almost identical by construction, so their BER and sum-rate curves overlap (Section 1).
2. BER and sum-rate rarely separate clustering rules in your system, because the cross-AP detector re-reads the non-serving APs whenever the cluster estimate is unreliable [1, 2]. A worse cluster costs *fronthaul*, not BER.
3. Metrics that *do* separate rules: fronthaul, gate firing rate, captured information, energy efficiency, tail SE and fairness, and above all the fronthaul needed to reach a target BER (Section 3).
4. Proposed: a **detector-aware information-surplus clustering** with a closed-form fronthaul-optimal cluster size, with a bound that links the cluster to Δ_k of your papers (Section 4).
5. Show the gain in an **interference-limited** regime, at **equal fronthaul** or **equal BER** (Section 5).

Citation conventions, verification status and what changed in this version

- “[x, eq. (y)]” or “[x, Sec. z]” points to the exact place in the arXiv version listed in the bibliography. “**Derived here**” or “**proposed here**” marks results that are not taken from a paper.
- The full texts of [10–13] were read and checked. All other references were checked at the level of title, authors and abstract (Table 2).
- **Correction in this version:** version 1 said that the information-rate rule of Mashdour and de Lamare includes inter-user interference. It does not: the rate in [10, eq. (11)] contains only the estimation error and noise. Section 1 is corrected accordingly.

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1 What your code does today (audit)

1.1 Channel-norm (CN) cluster

For each user k the code uses a metric and a threshold per propagation regime (code inspection, done here):

$$m_{l,k} = \begin{cases} \|\mathbf{h}_{l,k}^{\text{NF}}\|^2 \approx N\beta_{l,k}, & A_{l,k} = 1, \\ \beta_{l,k}, & A_{l,k} = 0, \end{cases} \quad D_{l,k}^{\text{CN}} = \mathbb{1}\left[m_{l,k} \geq \frac{1}{c} \max_{l' \in \mathcal{K}_k^{\text{reg}}} m_{l',k}\right], \quad (1)$$

with $c = \text{clu.div} = 2$, plus the master AP $\arg \max_l m_{l,k}$. A threshold on channel gain with a master AP follows the dynamic cooperation clustering (DCC) idea of [3].

Issue 1 (inconsistent scale). A near-field metric is about $N\beta$ and a far-field metric is β , so the master AP is almost always a near-field AP even when a far-field AP has a larger β . Use the same scale for both regimes.

1.2 Information-rate (IR / “BSR”) cluster

What the paper does. Mashdour and de Lamare define, for AP m and user k (downlink model with precoder $\mathbf{p}_k$, estimate $\hat{g}_{km}$ and error $\tilde{g}_{km}$) [10, eq. (11)]

$$\text{SR}_{km} = \log_2 \left(1 + \frac{\rho_f |\hat{g}_{km}|^2 \mathbf{p}_k^H \mathbf{p}_k}{\rho_f |\tilde{g}_{km}|^2 \mathbf{p}_k^H \mathbf{p}_k + \sigma_w^2} \right), \quad (2)$$

a network-wide threshold equal to the *average* rate [10, eq. (12)]

$$s_{rc} = \frac{1}{KM} \sum_{m=1}^M \sum_{k=1}^K \text{SR}_{km}, \quad (3)$$

and the cluster [10, eq. (13), Algorithm 1]

$$\mathcal{U}_k = \{m : \text{SR}_{km} \geq s_{rc}\} \cup \arg \max_m \text{SR}_{km}. \quad (4)$$

The rate in (2) has *no inter-user interference term*; it depends on the estimated gain, the estimation error and the noise, and therefore includes small-scale fading [10, Sec. III-B].

What your code does. It scores each link by

$$\text{SR}_{l,k} = \log_2 \left(1 + \frac{p(\text{tr} \mathbf{R}_{l,k} - \text{tr} \mathbf{C}_{l,k})}{p \text{tr} \mathbf{C}_{l,k} + 1} \right), \quad (5)$$

which has the same structure as (2) (signal over estimation error plus noise), but in statistical form. It then *ranks* APs by this score and keeps *exactly as many APs as the CN cluster has*.

Issue 2 (size rule differs from the paper). The paper’s cluster size comes from the average-rate threshold [10, eqs. (12)–(13)]; the code copies $|\mathcal{S}_k^{\text{CN}}|$ instead. With orthogonal pilots and good estimates, $\text{tr} \mathbf{C} \approx 0$, so the score is monotone in $\text{tr} \mathbf{R}_{l,k} \approx N\beta_{l,k}$ and ranks APs exactly like CN. With the same size and the same ranking, the clusters coincide; the code’s console line “AP-selection disagreement: CN vs Rate” reports this.

Issue 3 (shared limitation). Neither (2) nor the code score accounts for interference from other users, or for what the detector does with the cluster. This is the opening for the proposals in Section 4.

Consequence

To compare fairly with [10], implement its average-rate threshold (eqs. (12)–(13)) instead of the CN size. Any new rule must choose its own size and use information that CN does not use (interference or detector behaviour); otherwise it will again coincide with CN.

1.3 Why BER and sum-rate hardly move

- **The detector compensates.** When the reliability test fires, cross-AP list SIC fuses all APs [2, Sec. III-B], at an extra per-user fronthaul cost $\bar{B}_k = |\mathcal{S}_k| + \rho(L - |\mathcal{S}_k|)$ [2, Sec. III-B]. A weaker cluster makes the gate fire more often: the cost appears as fronthaul, not as BER.
- **Noise-limited regime.** With $K = 8$, $N = 64$ and $L = 8$, the local MMSE combiners suppress most interference; an earlier GBSE test in this project gave clusters of about 1.3 APs and no visible gain.
- **Shared combiner.** SIC, list-SIC and cross-AP share the same combiner, so the combiner-SINR sum rate cannot separate them.

The GBSA study reaches a related conclusion from the energy side: replacing its energy-efficiency objective by sum-rate or channel-norm objectives gives clearly lower energy efficiency at the same cluster-size limit [12, Sec. VI, Fig. 2]. The choice of *metric* decides what a clustering rule appears to achieve.

2 Literature review

Table 1: Clustering and AP-selection work relevant to you.

Work	Idea	Relevance / gap
Björnson, Sanguinetti [3]; Demir <i>et al.</i> [4]	Dynamic cooperation clustering, master AP, pilot assignment; scalable by construction.	The standard CN baseline.
Buzzi, D’Andrea [5]	User-centric cell-free: each user served by its strongest APs.	Baseline idea.
Ngo <i>et al.</i> [6]	AP selection to improve total energy efficiency with a power model that includes fronthaul.	EE power model.
Dao, Kim [7]	Effective-channel-gain metrics from large-scale fading; sequential association.	Large-scale only.
Ammar, Adve, Valaee, Yu [8]	Survey of user-centric cell-free networks.	Background.
Mashdour, Salehi, de Lamare, Schmeink, Lima [9]; Mashdour, de Lamare [10]	Boosted sum-rate (BSR) clustering: per-link rate with estimation error and noise [10, eq. (11)], average-rate threshold [10, eqs. (12)–(13)]; fairness scheduling.	Your IR baseline; no interference term; code uses a different size rule (Issue 2).
Tesolin, de Lamare [11]	Serving states as a graph with Hamming-distance adjacency; the number of candidate sets per user is $2^{D_k} - 1$ and of serving states $\prod_k (2^{D_k} - 1)$ [11, eqs. (1)–(2)]; graph-based search-and-evaluate (GBSE) with RAN energy efficiency [11, Sec. IV]; evaluated by processing time and optimality gap vs. exhaustive search [11, Sec. IV-B, Fig. 1].	An optimizer you can reuse with a new objective.

Work	Idea	Relevance / gap
Tesolin, de Lamare [12]	RAN energy efficiency (RANEE) objective [12, eq. (5)]; graph-based steepest ascent (GBSA) with fractional-programming power allocation and linear per-iteration complexity [12, Abstract, Secs. IV–V]; near-optimal RANEE vs. exhaustive search and a joint-optimisation baseline [12, Sec. VI, Fig. 3].	Gives the metrics used by your group.
Lai, Xiang, Lukito, Phan, Cheng, Liu, Mao [13]	HybridUA: online association that labels users network- or user-centric with a z-score of summed large-scale fading [13, Sec. III]; saves 71–94% of fronthaul signalling and up to 83% of CPU processing power with at most 8.6% SE loss; reports Jain’s index and 5% outage [13, Sec. IV].	A fronthaul-first evaluation to imitate.
Göttsch <i>et al.</i> [14, 15]	Fixed clusters under fairness scheduling; fronthaul quantization and routing.	Fairness and fronthaul modelling.
Ranasinghe <i>et al.</i> [16]; Pereira-Ruisánchez <i>et al.</i> [17]	Graph neural networks for AP selection and cluster formation.	Learning-based alternative.
Ssettumba <i>et al.</i> [18, 19]	Interference cancellation and IDD for clustered cell-free; AP selection with LLR refinement.	Closest to your detector side.
Wang <i>et al.</i> [20]; Mylonopoulos <i>et al.</i> [21]	Near-field cell-free combining and beam-focusing.	No near-field-specific clustering.

Gap. Within this search, no clustering rule was found that (a) is designed around a list/cross-AP detector and its reliability gate, or (b) uses near-field range-domain separability (estimated angle and range) to pick APs.

3 Metrics that prove a clustering is better

$\mathcal{S}_k$ is the cluster of user k , $\gamma_{l,k}$ the per-AP post-combining SINR (eq. (23) of [2]), $\Gamma_k = \sum_{l=1}^L \gamma_{l,k}$ and $\Gamma_{\mathcal{S}_k} = \sum_{l \in \mathcal{S}_k} \gamma_{l,k}$.

M1. Fronthaul per user (from [2, Sec. III-B]):

$$\bar{B} = \frac{1}{K} \sum_{k=1}^K (|\mathcal{S}_k| + \rho_k (L - |\mathcal{S}_k|)), \quad (6)$$

with ρ_k the measured gate firing rate (`etaXap_acc` in your code). Fronthaul is the primary metric in [13].

M2. Iso-performance cost (proposed here): the fronthaul $\bar{B}$ or energy needed to reach a target BER, e.g. 10^{-3} .

M3. Cluster size $|\mathcal{S}_k|$: mean and histogram, split into near- and far-field-dominated users. Average cluster size is also reported in [12, Fig. 3].

M4. Captured-information ratio (proposed here) $\kappa_k = \Gamma_{\mathcal{S}_k} / \Gamma_k$, and the information surplus Δ_k (eq. (26) of [2]). Plot their CDFs.

M5. Gate firing rate ρ vs. SNR (already measured by your code).

M6. Tail performance: 5th-percentile per-user SE (5% outage), as in [13].

M7. Fairness: Jain's index [24], also used in [13],

$$J = \frac{(\sum_k \text{SE}_k)^2}{K \sum_k \text{SE}_k^2} \in \left[\frac{1}{K}, 1\right]. \quad (7)$$

M8. Energy efficiency. GBSA defines RANEE as the sum rate over $LP_{\text{circuit}} + \sum_l p_l$, where P_{circuit} covers baseband, fronthaul, synchronisation and cooling [12, eq. (5)]. To make the fronthaul effect of your detector visible, split out a per-scalar fronthaul term (proposed here):

$$\text{EE} = \frac{B_w \sum_k \text{SE}_k}{L P_{\text{circ}} + \sum_l p_l + P_{\text{FH}} K \bar{B}}. \quad (8)$$

M9. Complexity: AP-side combiner cost, CPU-side LSFD cost $\sum_k \mathcal{O}(|\mathcal{S}_k|^3)$, and the run time of the clustering algorithm, as in [11, Fig. 1] and [12, Fig. 3].

M10. Load balance: $\max_l |\mathcal{D}_l| / (\frac{1}{L} \sum_l |\mathcal{D}_l|)$; users per CPU is reported in [13, Sec. IV].

M11. Robustness to CSI error: ratio of a metric under estimated vs. perfect CSI, per clustering rule.

Which ones to put in the paper

M1, M2, M4 (CDF of κ_k or Δ_k), M6 and M8. Make M2 the headline figure: BER vs. fronthaul at fixed SNR, one curve per clustering rule.

4 Proposed clustering strategies

4.1 P1: Detector-aware information-surplus clustering (main proposal, proposed here)

Choose, for each user, the *smallest* set of APs that captures a fraction $1 - \varepsilon$ of the user's total post-combining SINR, computed from the estimates of all AP-UE pairs (available after the cross-AP estimation of [2]) and including interference:

$$\hat{\gamma}_{l,k} = \frac{p |\hat{\mathbf{v}}_{l,k}^H \hat{\mathbf{h}}_{l,k}|^2}{p \sum_{j \neq k} |\hat{\mathbf{v}}_{l,k}^H \hat{\mathbf{h}}_{l,j}|^2 + \hat{\mathbf{v}}_{l,k}^H (p \sum_j \mathbf{C}_{l,j} + \mathbf{I}) \hat{\mathbf{v}}_{l,k}}, \quad \mathcal{S}_k = \arg \min_{\mathcal{S}} \left\{ |\mathcal{S}| : \sum_{l \in \mathcal{S}} \hat{\gamma}_{l,k} \geq (1 - \varepsilon) \sum_{l=1}^L \hat{\gamma}_{l,k} \right\}. \quad (9)$$

The SINR form follows eqs. (23)–(24) of [2] with the estimation-error covariance added; the minimum is found by sorting $\hat{\gamma}_{l,k}$ ($\mathcal{O}(L \log L)$ per user). Unlike (2), the score contains the interference left by the combiner.

Proposition 1 (Cluster target bounds the information surplus; derived here). *If $\kappa_k = \Gamma_{\mathcal{S}_k} / \Gamma_k \geq 1 - \varepsilon$, then the surplus of eq. (26) of [2] satisfies*

$$\Delta_k = \frac{\Gamma_k - \Gamma_{\mathcal{S}_k}}{(1 + \Gamma_{\mathcal{S}_k})(1 + \Gamma_k)} \leq \frac{\varepsilon}{1 + (1 - \varepsilon)\Gamma_k}. \quad (10)$$

Proof. $\Gamma_k - \Gamma_{\mathcal{S}_k} \leq \varepsilon \Gamma_k$ and $1 + \Gamma_{\mathcal{S}_k} \geq 1 + (1 - \varepsilon)\Gamma_k$, so $\Delta_k \leq \varepsilon \Gamma_k / ((1 + (1 - \varepsilon)\Gamma_k)(1 + \Gamma_k))$, and $\Gamma_k / (1 + \Gamma_k) \leq 1$. $\square$

One parameter ε controls how much the cluster-only estimate can lose against the all-AP estimate, which links clustering to the detector analysis of [1, 2]. The cluster size adapts per user, which is where fronthaul savings come from.

4.2 P2: Fronthaul-optimal cluster size for the gated cross-AP detector (proposed here)

The cluster scalars are always forwarded, and the other $L - |\mathcal{S}_k|$ only when the gate fires [2, Sec. III-B]. A larger cluster costs more base fronthaul but makes the gate fire less.

Gate model (derived here). The code declares a decision unreliable when the soft estimate is farther than $d_{\text{abs}} = d_{\text{th}} d_{\text{min}}/2$ from every constellation point (the shadow-region test, eq. (20) of [2]). If the fused error is $\mathcal{CN}(0, \varepsilon_{\mathcal{S}}^2)$ with $\varepsilon_{\mathcal{S}}^2 = 1/(1 + \Gamma_{\mathcal{S}})$ (eq. (25) of [2]), the probability of staying inside the disk around the true point is $1 - e^{-d_{\text{abs}}^2/\varepsilon_{\mathcal{S}}^2}$, so

$$\rho_k(\mathcal{S}) \approx \exp(-d_{\text{abs}}^2(1 + \Gamma_{\mathcal{S}})). \quad (11)$$

Optimal size. With APs sorted by $\hat{\gamma}_{l,k}$ and $\mathcal{S}^{(m)}$ the m best,

$$m_k^* = \arg \min_{m \in \{1, \dots, L\}} \left[m + \exp(-d_{\text{abs}}^2(1 + \Gamma_{\mathcal{S}^{(m)}}))(L - m) \right], \quad (12)$$

a search over L values per user. P2 picks the size, P1's ranking picks the APs. The model can be validated against the measured `etaXap_acc`.

4.3 P3: Near-field separability-aware ranking (proposed here)

In the near field an AP is valuable for user k if it can separate k from its strongest interferers in angle *or range*. Polar-domain estimation gives estimated angles and ranges for every pair [22], so the CPU can predict

$$\hat{\mu}_l(k, j) = |\mathbf{b}(\hat{\theta}_{l,k}, \hat{r}_{l,k})^H \mathbf{b}(\hat{\theta}_{l,j}, \hat{r}_{l,j})|, \quad \text{score}_{l,k} = \hat{\gamma}_{l,k} \left(1 - \max_{j \in \mathcal{I}_k} \hat{\mu}_l(k, j)^2 \right), \quad (13)$$

with $\mathcal{I}_k$ the strongest co-channel users of k . Your simulations create collinear user pairs, where this should help most. With sparse APs, $\hat{\mu}$ also reflects grating-lobe ambiguities; note that an aliased angle estimate leaves the channel vector correct but makes $\hat{\mu}$ wrong [23], so use the channel vectors directly when angles may alias.

4.4 P4: Graph-based search with the new objective

Use the GBSE/GBSA framework [11, 12] as the optimizer, with an objective that includes the detector (proposed here):

$$J(\{\mathcal{S}_k\}) = \sum_k \text{SE}_k - \lambda K \bar{B} \quad \text{or} \quad \text{EE (M8) with the gate term}, \quad (14)$$

started from the P1/P2 solution. GBSE searches Hamming neighbourhoods of the serving-state graph [11, Sec. III], and GBSA reaches near-optimal RANEE with linear per-iteration complexity [12]; the warm start should reduce the number of iterations.

4.5 Recommendation

What to propose

P1 + P2 as the proposed method (analytical, $\mathcal{O}(L \log L)$ per user, with Proposition 1 and the gate model), **P3** as the near-field enhancement, and **P4** as the near-optimal benchmark. A GNN version [16, 17] can be future work.

Algorithm 1 — P1+P2: detector-aware clustering (per coherence block, at the CPU)

Input: estimates $\hat{\mathbf{h}}_{l,k}$, error covariances $\mathbf{C}_{l,k}$, power p , target ε , gate radius d_{abs} .

1. For every AP l : compute the local MMSE combiners $\hat{\mathbf{v}}_{l,k}$ and the SINRs $\hat{\gamma}_{l,k}$ from (9).
2. For every user k :
 - (a) sort the APs so that $\hat{\gamma}_{l_1,k} \geq \hat{\gamma}_{l_2,k} \geq \dots$, and form $\Gamma^{(m)} = \sum_{i \leq m} \hat{\gamma}_{l_i,k}$;
 - (b) P1: $m_\varepsilon = \min\{m : \Gamma^{(m)} \geq (1 - \varepsilon)\Gamma^{(L)}\}$;
 - (c) P2: m^* from (12);
 - (d) $\mathcal{S}_k = \{l_1, \dots, l_{\max(m_\varepsilon, m^*)}\}$ (use min for a fronthaul-first design).

Cost: $\mathcal{O}(L \log L)$ per user after the combiners, which the receiver computes anyway.

5 How to show the gain (evaluation design)

Baselines. Single-AP; CN/DCC [3]; BSR with its own threshold rule [10, eqs. (11)–(13)]; effective-gain AP selection [7]; GBSE/GBSA with RANEE [11, 12]; full cooperation.

Regime. Clustering only matters when interference matters: use at least one of more users ($K = 16$ – 32), fewer antennas per AP ($N = 16$ – 32), collinear pairs, sparse APs, or estimated CSI. GBSA, for example, uses a small network (6 APs, up to 5 users) to compare with the exhaustive optimum [12, Sec. VI]; HybridUA varies the number of users, CPUs and the area [13, Sec. IV].

Figures. (1) BER vs. fronthaul $\bar{B}$ at fixed SNR (headline); (2) $\bar{B}$ and ρ vs. SNR, with the P2 model vs. measurement; (3) CDFs of κ_k and Δ_k (validates Proposition 1); (4) per-user SE CDF, 5% outage and Jain’s index; (5) cluster-size histogram, near- vs. far-field users; (6) EE vs. maximum cluster size, as in [12, Fig. 2]; (7) clustering run time vs. K .

6 Implementation notes for your MATLAB code

1. Compute clusters after channel estimation in each SNR loop, from $\hat{\mathbf{h}}$ and $\mathbf{C}$. With `genie_csi = false` the estimates are authentic; never use `h_det_all` in a clustering rule for an estimated-CSI result.
2. Do not tie the cluster size to `D_CN`; implement the average-rate threshold of [10, eqs. (12)–(13)] for the IR baseline.
3. Fix the master-AP scale (Issue 1).
4. Reuse `etaXap_acc` (ρ), `loadBSR_acc` (size) and `mBERrt*` (per-user BER); add κ_k , Δ_k and Jain’s index.
5. Run all rules on the same channel and noise realizations.

Honest risks

- In the default regime all rules may give the same BER; show the gain as fronthaul and EE at equal BER.
- The gate model in P2 assumes a Gaussian fused error and ignores the disks around other points; validate it against the measured rate.
- P3 depends on the accuracy of estimated angles and ranges.
- The BSR analysis in [10] uses a downlink model; the published letter [9] was not available here and may differ in details.

Verification status of the references

Table 2: What was checked for each reference.

References	Checked against
[10–13, 20, 23]	Full text.
[1, 2]	Your submitted papers.
[3, 4, 9, 18, 19, 22]	As cited in your papers.
[5–8, 14–17, 21, 24]	Title, authors and abstract only (web search).

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